Linux Commands for DevOps - Quick Reference Guide

1. Navigation Commands

Command	Example	Purpose
`pwd`	`pwd`	Show current directory path
`ls`	`ls`	List files and directories
`ls -l`	`ls -l`	Detailed file listing
`ls -la`	`ls -la`	Show hidden files with details
`cd directory`	`cd project`	Change directory
`cd ..`	`cd ..`	Move one directory up
`cd ~`	`cd ~`	Move to home directory

2. File & Directory Management

Command	Example	Purpose
`mkdir`	`mkdir project`	Create directory
`mkdir -p`	`mkdir -p app/logs`	Create nested directories
`touch`	`touch file.txt`	Create empty file
`cp`	`cp file.txt backup.txt`	Copy file
`cp -r`	`cp -r project backup`	Copy directory
`mv`	`mv file.txt newfile.txt`	Move/Rename file

`rm`	`rm file.txt`	Delete file
`rm -r`	`rm -r project`	Delete directory
`rm -rf`	`rm -rf project`	Force delete directory

3. File Viewing Commands

Command	Example	Purpose
`cat`	`cat file.txt`	Display file content
`less`	`less file.txt`	View large files page by page
`head`	`head file.txt`	Show first 10 lines
`tail`	`tail file.txt`	Show last 10 lines
`tail -f`	`tail -f app.log`	Monitor log file in real time
`nano`	`nano file.txt`	Edit file
`vim`	`vim file.txt`	Advanced text editor

4. Search Commands

Command	Example	Purpose
`find`	`find /home -name "*.txt"`	Find files
`grep`	`grep "error" app.log`	Search text in file
`grep -i`	`grep -i "error" app.log`	Case-insensitive search
`which`	`which python`	Find command location
`locate`	`locate nginx.conf`	Quickly locate files

5. User Management

Command	Example	Purpose
`useradd`	`sudo useradd ram`	Create user
`adduser`	`sudo adduser ram`	Create user interactively
`passwd`	`sudo passwd ram`	Set user password
`id`	`id ram`	View user details
`whoami`	`whoami`	Show current user
`su -`	`su - ram`	Switch user
`userdel`	`sudo userdel ram`	Delete user
`userdel -r`	`sudo userdel -r ram`	Delete user and home directory

Group Management & Access Control

Group Commands

Command	Example	Purpose
`groupadd`	`sudo groupadd developers`	Create group
`groupdel`	`sudo groupdel developers`	Delete group
`groups`	`groups ram`	View group membership
`usermod -aG`	`sudo usermod -aG developers ram`	Add user to group
`gpasswd -d`	`sudo gpasswd -d ram developers`	Remove user from group
`getent group`	`getent group developers`	View group details

Permission Commands

Command	Example	Purpose
`ls -l`	`ls -l`	View permissions
`chmod`	`chmod 755 file.sh`	Change permissions
`chmod +x`	`chmod +x script.sh`	Make executable
`chown`	`chown ram file.txt`	Change owner
`chown user:group`	`chown ram:developers file.txt`	Change owner and group
`chgrp`	`chgrp developers file.txt`	Change group ownership

Permission Numbers

Number	Permission	Meaning
7	rwx	Read, Write, Execute
6	rw-	Read, Write
5	r-x	Read, Execute
4	r--	Read Only
0	---	No Permission

Common Permission Examples

Command	Meaning
`chmod 777 file.txt`	Full access to everyone
`chmod 755 script.sh`	Owner full, others read & execute
`chmod 644 file.txt`	Owner read/write, others read
`chmod 700 file.txt`	Owner only access

`chmod 770 folder`	Owner & group full access

Practical DevOps Access Control Example

Scenario

Users: Ram, Sham, Jay

Group: developers

Shared Directory: /project

Step 1 - Create Users

Command
`sudo useradd ram`
`sudo useradd sham`
`sudo useradd jay`
`sudo passwd ram`
`sudo passwd sham`
`sudo passwd jay`

Step 2 - Create Group

Command
`sudo groupadd developers`

Step 3 - Add Users to Group

Command

```
sudo usermod -aG
developers ram
```

```
sudo usermod -aG
developers sham
```

Jay is not added to the group.

Step 4 - Create Shared Directory

Command

```
sudo mkdir
/project
```

Step 5 - Assign Group Ownership

Command

```
sudo chown root:developers
/project
```

Step 6 - Give Permissions

Command

```
sudo chmod 770
/project
```

Access Result

User	Group Member	Access to /project
Ram	Yes	Allowed
Sham	Yes	Allowed
Jay	No	Denied

Process Management

Command	Example	Purpose
`ps`	`ps -ef`	View running processes
`top`	`top`	Real-time process monitoring
`htop`	`htop`	Interactive process viewer
`kill`	`kill 1234`	Stop process
`kill -9`	`kill -9 1234`	Force kill process
`pkill`	`pkill nginx`	Kill process by name

Networking Commands

Command	Example	Purpose
`ip addr`	`ip addr`	Show IP address
`hostname -I`	`hostname -I`	Show server IP
`ping`	`ping google.com`	Test connectivity
`curl`	`curl google.com`	Fetch web content
`wget`	`wget file.zip`	Download files
`nslookup`	`nslookup google.com`	DNS lookup
`dig`	`dig google.com`	Detailed DNS lookup
`ss -tulnp`	`ss -tulnp`	Check listening ports
`netstat -tulnp`	`netstat -tulnp`	View open ports

Disk & Memory Commands

Command	Example	Purpose
`df -h`	`df -h`	Disk usage
`du -sh`	`du -sh folder`	Folder size
`free -h`	`free -h`	Memory usage
`lsblk`	`lsblk`	Disk information
`mount`	`mount`	Mounted filesystems

Archive & Compression

Command	Example	Purpose
`tar -cvf`	`tar -cvf backup.tar project`	Create archive
`tar -xvf`	`tar -xvf backup.tar`	Extract archive
`tar -czvf`	`tar -czvf backup.tar.gz project`	Compress archive
`tar -xzvf`	`tar -xzvf backup.tar.gz`	Extract compressed archive
`zip`	`zip backup.zip file.txt`	Create zip
`unzip`	`unzip backup.zip`	Extract zip

Service Management (Very Important for DevOps)

Command	Example	Purpose
`systemctl status`	`systemctl status nginx`	Check service status
`systemctl start`	`systemctl start nginx`	Start service
`systemctl stop`	`systemctl stop nginx`	Stop service
`systemctl restart`	`systemctl restart nginx`	Restart service
`systemctl reload`	`systemctl reload nginx`	Reload config
`systemctl enable`	`systemctl enable nginx`	Start at boot
`systemctl disable`	`systemctl disable nginx`	Disable at boot

Log Management

Command	Example	Purpose
`journalctl -xe`	`journalctl -xe`	View system logs
`journalctl -u nginx`	`journalctl -u nginx`	Service logs
`tail -f`	`tail -f /var/log/messages`	Live log monitoring
`dmesg`	`dmesg`	Kernel logs

Bash Scripting Basics

Command	Example
`chmod +x script.sh`	Execute script permission
`./script.sh`	Run script
`echo`	Print output
`read`	Take user input
`if`	Conditional statements
`for`	Loop execution
`while`	Repeated execution